

Ion dissociation constants of weak acids and bases (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.1.9) Interconvert $[H^+]$, $[OH^-]$, pH and pOH values.
- (R3.1.10) Write K_a and K_b expressions.
- (R3.1.10) Interpret the relative strengths of acids and bases from the K_a , K_b , pK_a or pK_b values.

In [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>), we saw the difference between strong and weak acids and bases. We have seen that the pH of equimolar solutions of weak and strong acid solutions are different, with the weak acid having a higher pH. But how can we quantify this behaviour and how can the pH of a weak acid or base be calculated knowing an equilibrium is established? In [section R3.1.4–5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)), you learned about the relationship between H^+ and OH^- , but solutions are typically expressed in terms of pH, which is based on the hydrogen ion concentration. What about the hydroxide concentration? Can we express pOH, just like we express pH?

pOH calculations

We have previously seen that pH indicates the acidity of a solution, and it is related to $[H^+]$ through the logarithmic equation:

$$pH = -\log[H^+]$$

Similarly, to indicate basicity, the pOH of a solution can be calculated by taking the negative log of the hydroxide ion concentration, $[OH^-]$, using the following equation:

Table 1. pOH equation.

Equation	Symbols	Units
$pOH = -\log[OH^-(aq)]$	pOH = a measure of how basic a species is	No units
	$[OH^-(aq)] =$ concentration	Moles per cubic decimetre dm^{-3}

Equation	Symbols	Units
	of aqueous hydroxide ion	

We can also derive the formula for the concentration of hydroxide ions using pOH:

$$[\text{OH}^-] = 10^{-\text{pOH}}$$

Recall from [section R3.1.4–5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)), that:

$$[\text{H}^+] \times [\text{OH}^-] = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

We can therefore interconvert between $[\text{H}^+]$, $[\text{OH}^-]$, pH and pOH.

Study skills

The equation for pOH is given in the DP chemistry data booklet in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>).

Worked example 1

For a solution with $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$ calculate the:

- pOH
- pH

1.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{OH}^-] = 1.0$
Step 2: Write out the equation.	$\text{pOH} = -\log$
Step 3: Substitute the values given.	$= -\log(1.0)$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= 4$ Note that pO same reason

2.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{OH}^-] = 1.0$
Step 2: Write out the required equation.	$[\text{OH}^-] \times [\text{H}^+]$
Step 3: Rearrange the equation.	$[\text{H}^+] = \frac{1 \times}{[\text{O}]}$
Step 4: Substitute the values given.	$= \frac{1 \times 10^{-1}}{1 \times 10^{-4}}$ $= 1 \times 10^{-10}$
Step 5: Write out the required equation.	$\text{pH} = -\log[$
Step 6: Substitute the values calculated.	$\text{pH} = -\log[$
Step 7: State the answer with appropriate units and the number of significant figures used in rounding.	$= 10$

Try these questions for yourself.



Exercises 1 and 2

Click a question to answer



1. True or false?



A solution with $\text{pOH} = 6$ has $[\text{H}^+] = 1 \times 10^{-6} \text{ mol dm}^{-3}$.



2. True or false?



A solution with $[\text{H}^+] = 1 \times 10^{-4} \text{ mol dm}^{-3}$ has $\text{pOH} = 10$.

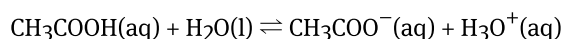


Check answers

Higher level (HL)

Acid and base dissociation constants

Strong acids and alkalis completely dissociate (or ionise) into ions in aqueous solution and as such the pH of solutions of strong acids and bases is relatively straightforward to calculate as seen in [section R3.1.4-5a](#) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)). However, as seen in [section R3.1.6a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>), there are a wide range of acids and bases where the degree of dissociation or ionisation is not complete. Such acids and bases are referred to as weak acids and bases - there exists an equilibrium between the dissociated and undissociated forms. The following equation shows the equilibrium for a solution of ethanoic acid, CH₃COOH.

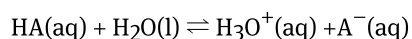


Weak acids such as ethanoic acid partially dissociate in solution. Unlike strong acids and strong bases, we cannot determine the [H⁺] or [OH⁻] from their initial concentrations. This is because each weak acid or weak base has a different degree of dissociation.

Just like equilibrium reactions, the dissociation of a weak acid or weak base has a constant associated with it as seen in [section R2.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/>), which tells us the extent of this dissociation and where the position of equilibrium lies as seen in [section R2.3.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-reaction-quotient-q-hl-id-46068/>). For weak acids, this is the acid dissociation constant, K_a .

The acid dissociation constant, K_a

The dissociation of a generic weak acid, HA, can be represented as:



The expression for the acid dissociation constant, K_a , for the dissociation reaction is given in **Tables 2a** and **2b**.

Table 2a. Acid dissociation constant.

Equation	Symbols	Units
$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$	K_a is the acid dissociation constant	Moles per cubic deci (mol dm ⁻³)
	[HA] is the concentration of the acid.	Moles per cubic deci (mol dm ⁻³)
	[H ₃ O ⁺] is the concentration of the hydronium ion.	Moles per cubic deci (mol dm ⁻³)

Equation	Symbols	Units
	$[A^-]$ is the concentration of the A^- ion.	Moles per cubic decimetre (mol dm^{-3})

or

Table 2b. Acid dissociation constant

Equation	Symbols	Units
$K_a = \frac{[H^+][A^-]}{[HA]}$	K_a is the acid dissociation constant	Moles per cubic decimetre (dm^{-3})
	$[HA]$ is the concentration of the acid.	Moles per cubic decimetre (dm^{-3})
	$[H^+]$ is the concentration of the hydrogen ion.	Moles per cubic decimetre (dm^{-3})
	$[A^-]$ is the concentration of the A^- ion.	Moles per cubic decimetre (dm^{-3})

Note that this reaction equation is an example of heterogeneous equilibrium because there are a mixture of states - both liquid and aqueous - in the reaction. Concentration is used to quantify mixtures, so the aqueous species are included in the equilibrium expression. However, water, is a pure substance and is therefore not expressed by concentration. Therefore, water is not included in the final equilibrium expression.

K_a is known as the acid dissociation constant (or acid ionisation constant). It has a fixed value for a particular acid at a particular temperature - it is temperature-dependent.

Worked example 2

Write the K_a expression for the dissociation of the weak acid, HF.

Solution steps	Calculations
Step 1: Write out the dissociation equation.	$\text{HF(aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$
Step 2: Write the K_a expression,	$K_a = \frac{[\text{H}_3\text{O}^+][\text{F}^-]}{[\text{HF}]}$

How do you think the K_a value and the strength of an acid are related?

As seen in subtopic R2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), the higher the extent of the reaction, the more favoured the products are. Therefore, the higher the concentrations of the products. For a weak acid, the reactants are favoured. The weaker the acid, the less it dissociates, and the lower the concentration the products and the higher the concentration of the acid, making the K_a ratio small. As such, the value of the K_a tells us about the extent of dissociation and therefore the strength of a weak acid and therefore the degree of dissociation. A weak acid with a higher K_a value is stronger and more dissociated in solution than a weak acid with a lower K_a value (**Figure 1**).

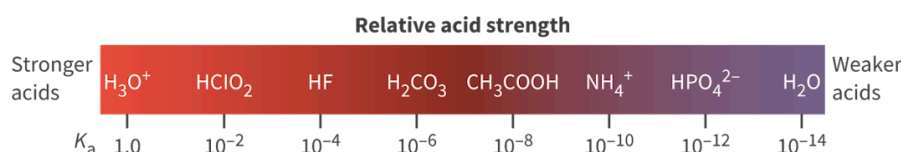


Figure 1. K_a values.

 More information for figure 1

This image is a diagram illustrating the relative acid strength using K_a values. The diagram is a horizontal bar with gradients transitioning from red to purple, indicating stronger to weaker acids. At the leftmost, the bar indicates 'Stronger acids' with a label for a K_a of 1.0 showing H_3O^+ . Progressing to the right, the K_a values decrease: $HClO_2$ (10^{-2}), HF (10^{-4}), H_2CO_3 (10^{-6}), CH_3COOH (10^{-8}), NH_4^+ (10^{-10}), HPO_4^{2-} (10^{-12}), and ending with H_2O at a K_a of 10^{-14} , which is the weakest. Above the bar, the text 'Relative acid strength' is written, providing context for the strength gradient depicted in the diagram.

[Generated by AI]

Worked example 3

Which acid is strongest? Which is the weakest?

Acid	Formula	K_a
Methanoic acid	$HCOOH$	1.8×10^{-4}
Ethanoic acid	CH_3COOH	1.8×10^{-5}
Propanoic acid	C_2H_5COOH	1.4×10^{-5}

The largest K_a value is 1.8×10^{-4} and therefore the strongest acid is methanoic acid.

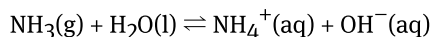
The lowest K_a value is 1.4×10^{-5} and therefore the weakest acid is propanoic acid.

Study skills

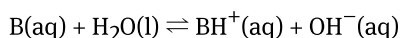
In section R1.4.4a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/>), we looked at the relationship between the equilibrium position and the Gibbs free energy change, ΔG . Students should be aware that the ΔG can also be calculated using the K_a or K_b for a weak acid or base (and vice versa) instead of the equilibrium constant, K_c .

The base dissociation constant, K_b

Ammonia, NH_3 , is an example of a weak base that partially dissociates in solution:



In a similar argument to that used above for weak acids, we can consider the ionisation of a weak base using the generic weak base B:



and derive an expression for K_b , the base dissociation constant.

Table 3. Base dissociated constant.

Equation	Symbols	Units
$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$	K_b is the base dissociation constant.	Moles per cubic deci (mol dm ⁻³)
	$[\text{B}]$ is the concentration of the base.	Moles per cubic deci (mol dm ⁻³)
	$[\text{BH}^+]$ is the concentration of the BH^+ ion.	Moles per cubic deci (mol dm ⁻³)
	$[\text{OH}^-]$ is the concentration of the hydroxide ion.	Moles per cubic deci (mol dm ⁻³)

K_b , the base dissociation constant, has a fixed value for a particular base at a specific temperature.

Worked example 4

Write the K_b expression for the dissociation of the weak base, CH_3NH_2 .

Solution steps	Calculations
Step 1: Write out the dissociation equation.	$\text{CH}_3\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$

Solution steps	Calculations
Step 2: Write the K_b expression.	$K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2]}$

As seen in subtopic R2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), the higher the extent of the reaction, the more favoured the products are. Therefore, the higher the concentrations of the products. For a weak base, the reactants are favoured. The weaker the base, the less it dissociates, and the lower the concentration of the products and the higher the concentration of the base, making the K_b ratio small. As such, Similar to the K_a , the value of K_b tells us about the strength of the base and therefore the degree of dissociation. A weak base with a high K_b value is stronger and more dissociated than a weak base with a low K_b value (**Figure 2**).

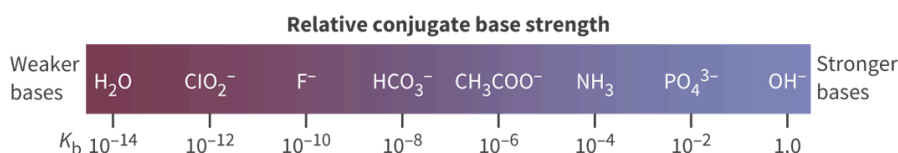


Figure 2. K_b values.

 More information for figure 2

The image is a diagram illustrating the relative conjugate base strength. It shows a gradient bar with the label 'Relative conjugate base strength' at the top. On the left, it mentions 'Weaker bases', and on the right, 'Stronger bases'. The diagram lists various bases along the bar, each associated with a specific K_b value, which indicates their strength. From left to right:

- H_2O with $K_b 10^{-14}$
- ClO_2^- with $K_b 10^{-12}$
- F^- with $K_b 10^{-10}$
- HCO_3^- with $K_b 10^{-8}$
- CH_3COO^- with $K_b 10^{-6}$
- NH_3 with $K_b 10^{-4}$
- PO_4^{3-} with $K_b 10^{-2}$
- OH^- with $K_b 1.0$

[Generated by AI]

Worked example 5

Which base is strongest and which is the weakest?

Base	K_b
A	1.8×10^{-4}
B	1.8×10^{-3}
C	1.4×10^{-6}

The largest K_b value is $1.8 \times 10^{-3} \text{ mol dm}^{-3}$ and therefore the strongest base is B.

The lowest K_b value is $1.4 \times 10^{-6} \text{ mol dm}^{-3}$ and therefore the weakest base is C.

Study skills

The values of K_a and K_b are constant for a specific acid and base respectively, at a specified temperature. K_a and K_b values tell us about the strength of a weak acid or weak base and the degree of dissociation.

These values will be given in the exam questions where needed.

pK_a and pK_b

Like we calculated pH from $[H^+]$ and pOH from $[OH^-]$ in order to convert numbers with exponents to easier numbers for comparison purposes, we can calculate pK_a from K_a and pK_b from K_b . Here are the formulas that you can use:

$$pK_a = -\log K_a \quad \text{and} \quad pK_b = -\log K_b$$

$$K_a = 10^{-pK_a} \quad \text{and} \quad K_b = 10^{-pK_b}$$

Look at the K_a , pK_a , K_b and pK_b values in **Table 4** and **Table 5**.

Can you find a relationship between strength of acid, K_a and pK_a , and between strength of base, K_b and pK_b ?

Table 4. The K_a and pK_a values of some weak acids.

Acid	Formula	K_a	
Methanoic acid	HCOOH	1.8×10^{-4}	
Ethanoic acid	CH ₃ COOH	1.8×10^{-5}	
Propanoic acid	C ₂ H ₅ COOH	1.4×10^{-5}	

Table 5. The K_b and pK_b values of some weak bases.

Base	Formula	K_b	p
Ammonia	NH_3	1.8×10^{-5}	4
Methylamine	CH_3NH_2	4.6×10^{-4}	3
Ethylamine	$\text{C}_2\text{H}_5\text{NH}_2$	4.5×10^{-4}	3

As the values above show, the stronger the acid, the larger the K_a and the smaller the pK_a . Similarly, the stronger the base, the larger the K_b and the smaller the pK_b (**Figure 3**).

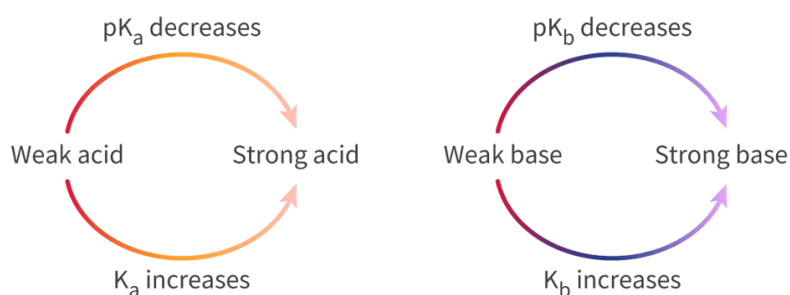


Figure 3. The relationship between K_a and pK_a with acid strength, and the equivalent relationship for bases.

 More information for figure 3

The image is a diagram with two circular arrows. The left circle represents acids and includes the sequence: 'Weak acid' at the bottom left, 'Ka increases' in clockwise direction at the bottom, 'Strong acid' at the top right, and 'pKa decreases' at the top. The right circle represents bases and includes the sequence: 'Weak base' at the bottom left, 'Kb increases' in clockwise direction at the bottom, 'Strong base' at the top right, and 'pKb decreases' at the top. The diagram visually demonstrates that as the strength of an acid increases, its K_a value increases and pK_a value decreases, and similarly for bases with K_b and pK_b .

[Generated by AI]

Worked example 6

The pK_b of a base is 3.34.

Calculate its K_b .

Solution steps

Step 1: Write out the values given in the question.

Solution steps

Step 2: Write out the equation.

Step 3: Substitute the values given.

Step 4: State the answer with appropriate units and the number figures used in rounding.

Worked example 7

The K_a of an acid is 1.8×10^{-5} .

Calculate its pK_a .

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Substitute the values given.

Step 4: State the answer with appropriate units and the number significant figures used in rounding.

Worked example 8

Looking at the values of K_a , pK_a , K_b and pK_b in **Table 4** above.

Which acid is stronger, methanoic acid or ethanoic acid?

Since the pK_a of methanoic acid is less than the pK_a of ethanoic acid, methanoic acid is stronger.

Worked example 9

Looking at the values of K_a , pK_a , K_b and pK_b in the **Table 5** above.

Which base is weaker, ammonia or methylamine?

Since the pK_b of methylamine is lower than the pK_b of ammonia,

ammonia is a weaker base than methylamine.

Use **Interactive 1** to consolidate your understanding of K_a and pK_a .

Drag the words into the correct boxes

1. The pK_a of an acid is 4, therefore the K_a is .
2. If an acid A has a pK_a value of 4, and an acid B a pK_a of 10, then A is than B.
3. If a base A has a pK_b value of 6 and a base B a pK_b of 2, then base A is than B.

1×10^{-4}

weaker

stronger

✓ Check

Interactive 1. The strength of some weak acids and bases.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions:

1. Go on a hunt! Find the K_a or K_b of at least eight materials found in real contexts.
2. Calculate their pK_a or pK_b .
3. Create a rating scale, arranging the materials by acid—base strength. Include the substance and its K_a or K_b and pK_a or pK_b .